MIL-L-45456E
6 December 1985
SUPERSEDING
MIL-L-45456D
27 October 1976

MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC BELT, 20MM: M17A2

This specification is approved for use by the all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of metallic belt link intended to hold 20 millimeter (mm) ammunition in the form of flexible belts for use in M61 type 20mm automatic weapons systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-13855	- Weapons, Small Arms, General Specification for
DOD-P-16232	- Phosphate, Coating, Heavy Manganese or Zinc Base
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1305

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

7268376	-	Pin
8448077	-	Loop, Long
8448078	-	Link, Cartridge, Metallic, Belt, 20mm: M17A2
P8448078	-	Packaging Data Sheet for Link, Cartridge, Metallic Belt, 20mm: M17A2
8448094	-	Loop, Short
9381628	-	Test Diagram and Requirements for Link, Cartridge, Metallic Belt, 20mm: M17A2

INSPECTION EQUIPMENT DRAWINGS

7273261	-	Snap Gage, Single End
7273262	-	Gage, Flat Plug (Not Go)
7273263	-	Gage, Flat Plug (Go)
7273417	-	Gage, Special Flat Plug
7273688	-	Gage, Flush Pin
7273733	-	Gage, Location
7274001	-	Gage, Test Fixture
7274003	-	Gage, Test Fixture
7274004	-	Gage, Test Mandrel
7274006	-	Gage, Test Fixture
7797882	-	Gage, Snap, Double End
7798081	-	Gage, Flush Pin
7798103	-	Gage, Flush Pin
11838148	-	Gage, Receiver
11838461	-	Gage, Snap, Double End

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E18 -	Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B117 -	Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components. The links shall comply with all requirements specified on drawings 8448078 and 9381628 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Detent retention. The links shall be capable of withstanding an applied force of 100 pounds as shown on Drawing 9381628 without the front gripping band detent disengaging from the cartridge. Testing shall be as specified in 4.5.4.

3.4 Tensile load. The links assembled into belts of 5 links with 20mm M51 dummy cartridges shall be capable of withstanding an 800 pound tensile load as shown on Drawing 9381628 without breakage or belt separation. Testing shall be as specified in 4.5.5.

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3.5 Cartridge stripping. The force required to strip Government standard 20mm dummy cartridges from links shall not be less than 220 pounds nor more than 275 pounds as shown on Drawing 9381628. Testing shall be as specified in 4.5.6.

3.6 Free hinging. Links assembled into belts of 25 links each using M51 series 20mm dummy cartridges shall hinge and fold over without binding, kinking, separating or any interference. The testing shall be performed as specified in 4.5.7.

3.7 Belt length. Links, when assembled into 25 round belts with Government standard 20mm dummy cartridges or steel mandrels Drawing 7274004, shall have a length within the limits specified on Drawing 9381628. Testing shall be as specified in 4.5.8.

3.8 Functioning. Links shall be capable of being assembled with Government standard 20mm cartridges into belts and fired through the 20mm ASG-21 (B-52 Tail Gun) Fire Control System or equivalent without cracking, breaking, separating or malfunctioning. Testing shall be as specified in 4.5.9.

3.9 Marking. The links shall be marked in accordance with Drawings 8448077 and 8448094. In addition, the links shall be identified with the inspection lot number as specified in MIL-STD-1168.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. Workmanship shall be in accordance with MIL-W-13855. Cracks, tears and burrs in non-principal areas and deformations shall not exceed the limits specified in 6.5.1, 6.5.2.2 and 6.5.3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (4.3)

b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as indicated.

Parts description	Drawings	Quantity
Link, Cartridge, Metallic Belt, 20mm: M17A2	8448078	2400
Loop, Short	8448094	263
Loop, Long	8448077	263
Pin	7268376	263

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. Rejection of the first article sample shall be in accordance with Table I and MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-L-45456E

PARAGRAPH	TITLE: Link, Cartridge, Metallic Belt, 20mm: M17A2		SHEET 1 OF 3		DRAWING NUMBER: SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin (Dwg 7268376)				
	Examination for defects	200	Acc-0 Rej-1	3.2	4.4.2.1
	Decarburization	5	Acc-0 Rej-1	3.2	4.5.1
	Salt Spray	8	Acc-0 Rej-1	3.2	4.5.2
	Hardness	50	Acc-0 Rej-1	3.2	4.5.3
	Loop, Short (Dwg 8448094)				
	Examination for defects	200	Acc-0 Rej-1	3.2	4.4.2.2
	Decarburization	5	Acc-0 Rej-1	3.2	4.5.1
	Salt Spray	8	Acc-0 Rej-1	3.2	4.5.2
	Hardness	50	Acc-0 Rej-1	3.2	4.5.3

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-L-45456E

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20mm: M17A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Loop, Long (Dwg 8448077)				
	Examination for defects	200	Acc-0 Rej-1	3.2	4.4.2.3
	Decarburization	5	Acc-0 Rej-1	3.2	4.5.1
	Salt Spray	8	Acc-0 Rej-1	3.2	4.5.2
	Hardness	50	Acc-0 Rej-1	3.2	4.5.3
	Link, Cartridge, Metallic Belt, 20mm: M17A2 (Dwg 8448078)				
	Examination for defects	200	Acc-0 Rej-1	3.2	4.4.2.4 1/
	Functioning	600	Acc-0 Rej-1	3.8	4.5.9

NOTES: 1/ May be included for the detent retention sample

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-L-45456E

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Link, Cartridge, Metallic Belt, 20mm: M17A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Test Diagrams and Requirements (for Link, Cartridge, Metallic Belt, 20mm: M17A2 (Dwg 9381628)				
	Detent Retention	500	Acc-2 Rej-3	3.3	4.5.4/7274003/ 7274004
	Tensile Load	200	Acc-0 Rej-1	3.4	4.5.5/7274001/ 7274004 1/
	Cartridge Stripping	500	Acc-4 Rej-5	3.5	4.5.6/7274006/ 7274004
	Free Hinging	800	Acc-0 Rej-1	3.6	4.5.7
	Belt Length	800	Acc-0 Rej-1	3.7	4.5.8/7274004 2/

NOTES: 1/ Selected from cartridge stripping test sample.
2/ Selected from free hinging test sample.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078 and MIL-W-13855. In addition, inspection lots shall not exceed 500,000 links.

4.4.1.1 Heat treat lot. The heat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one charge or in a continuous type of furnace during an uninterrupted period extending over not more than one ten hour period.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7268376
4.4.2.1	Pin (Link, 20mm, M17A2)				NEXT HIGHER ASSEMBLY 8448078
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR					
101.	Diameter, head		1.00%	3.2	Gage
102.	Diameter, shaft		1.00%	3.2	Gage
103.	Overall length		1.00%	3.2	Gage
	TEST				
104.	Decarburization	5 1/	Acc-0 Rej-1	3.2	4.5.1
105.	Salt spray	2/	2/	3.2	4.5.2
106.	Hardness	1/3/	40%	3.2	4.5.3
MINOR					
201.	Incorrect or missing protective finish		2.50%	3.2	Visual
202.	Evidence of poor workmanship		4.00%	3.11	Visual

NOTES:
1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Loop, Short (Link, 20mm, M17A2)	1 OF 2	8448094
			NEXT HIGHER ASSEMBLY: 8448078

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR					
101.	Cracks-tears or burrs in principal areas		.065%	3.2	Visual/6.5.2.1/6.5.3.1/ 6.5.4
102.	Diameter, outside flange (max)		.65%	3.2	11838148
103.	Diameter, inside (min)		.65%	3.2	11838148
104.	Width, inside "T" slot (min)		.65%	3.2	7273688
105.	Height, inside diameter to underside of "T' slot		.65%	3.2	7273688
106.	Opening, "T" slot		1.0%	3.2	7273262/7273263
107.	Length, overall		1.0%	3.2	Gage
	TESTS				
108.	Decarburization	5 1/	Acc-0 Rej-1	3.2	4.5.1
109.	Salt spray	2/	2/	3.2	4.5.2
110.	Hardness	1/3/	.40%	3.2	4.5.3

NOTES: 1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3

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CLASSIFICATION OF DEFECTS & TESTS

MIL-L-45456E

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 8448094
4.4.2.2	Loop, Short (Link, 20mm, M17A2)				NEXT HIGHER ASSEMBLY 8448078
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Missing or improper identification		1.50%	3.2	Visual
202.	Incorrect or missing protective finish		2.50%	3.2	Visual
203.	Evidence of poor workmanship		4.00%	3.11	Visual/6.5.1/6.5.2.2/ 6.5.3.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER 8448077
4.4.2.3	Loop, Long (Link, 20mm, M17A2)		NEXT HIGHER ASSEMBLY 8448078

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR					
101.	Cracks-tears or burrs principal areas		.065%	3.2	Visual/6.5.2.1/6.5.3.1/ 6.5.4
102.	Location, detents for cartridge crimping groove from nesting section		.15%	3.2	7798081
103.	Location, forward base of front gripping band from nesting section		.15%	3.2	7798081
104.	Height, vertical tab, front gripping band		.25%	3.2	7798103
105.	Length, front gripping band (max)		.65%	3.2	7273261
106.	Length, overall		.65%	3.2	Gage
107.	Height location, inside front and rear guide lugs from gage line		.65%	3.2	7273733
108.	Height location, rear cartridge bearing indent from gage line		.65%	3.2	7273733
109.	Location, outside high points, front and rear guide lugs, from gage line		.65%	3.2	7273733
110.	Location, nesting section from front guide lug.		.65%	3.2	7798103

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 8448077
4.4.2.3	Loop, Long (Link, 20mm, M17A2)				NEXT HIGHER ASSEMBLY 8448078
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111.	Width, nesting section		.65%	3.2	7273417
112.	Length location, vertical tab from nesting section		.65%	3.2	11838461
113.	Nesting flange to bottom of nesting surface		.65%	3.2	Gage
114.	Distance across nesting section		.65%	3.2	Gage
115.	Location, rear base of rear gripping band from front guide lug		1.00%	3.2	7798081
	TESTS				
116.	Decarburization	5 1/	Acc-0 Rej-1	3.2	4.5.1
117.	Salt spray	2/	2/	3.2	4.5.2
118.	Hardness	1/3/	.40%	3.2	4.5.3

NOTES: 1/ Selected from each heat treat lot (See 4.4.1.1)
2/ Sampling and rejection shall be in accordance with DOD-P-16232.
3/ Samples shall be in accordance with MIL-STD-105, Level S-3

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 8448077
4.4.2.3	Loop, Long (Link, 20mm, M17A2)				NEXT HIGHER ASSEMBLY 8448078
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Width, vertical tab, front gripping band		1.50%	3.2	7797882
202.	Missing or improper identification		1.50%	3.2	Visual
203.	Incorrect or missing protective finish		1.50%	3.2	Visual
204.	Nesting flange radii's		2.50%	3.2	Visual
205.	Radii tie-bar cutoff		2.50%	3.2	Visual
206.	Evidence of poor workmanship		4.00%	3.11	Visual/6.5.1/6.5.2.2/ 6.5.3.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8448078
4.4.2.4	Link, Cartridge Metallic Belt, 20mm, M17A2				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR					
101.	Improperly assembled parts		.15%	3.2	Visual
102.	Pin, flaring (min)		.15%	3.2	SMTE
MINOR	None defined.				

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Test Diagrams and Requirements for: Link, Cartridge, Metallic Belt, 20mm, M17A2				9381628
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR					
101.	Detent retention	500	Acc-5 Rej-6	3.3	4.5.4/7274003/7274004
102.	Tensile load	200	Acc-1 Rej-2	3.4	4.5.5/7274001/ 7274004 1/
103.	Cartridge stripping	500	Acc-5 Rej-6	3.5	4.5.6/7274006/7274004
104.	Free hinging	800	Acc-0 Rej-1	3.6	4.5.7
105.	Belt length	800	Acc-0 Rej-1	3.7	4.5.8/7274004 2/
MINOR	None defined.				

NOTES: 1/ Selected from cartridge stripping test sample.
2/ Selected from free hinging test sample.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Link, Cartridge, Metallic Belt, 20mm, M17A2, Packaging Data Sheet				P8448078
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
MAJOR	None defined.				
MINOR					
201.	Improper level of packaging and packing		1.50%	3.2	Visual
202.	Incorrect, missing or illegible markings on containers		1.50%	3.2	Visual
203.	Inadequate cleaning or drying of links		1.50%	3.2	Visual
204.	Unit container interior contaminated with dirt, lint, or other foreign matter		1.50%	3.2	Visual
205.	Improper orientation of links in unit container		1.50%	3.2	Visual
206.	Improper oriented link cartons in shipping container		1.50%	3.2	Visual
207.	Missing or improper separators between layers and top and bottom of link cartons		1.50%	3.2	Visual
208.	Incorrect quantity of links per shipping container		1.50%	3.2	Visual
209	Improper closure of shipping container		1.50%	3.2	Visual
210.	Missing or improper trays		1.50%	3.2	Visual

NOTES:

AM[illegible] Form 1570, 1 Feb 85 Re[illegible]es DRSMC-QA (D) Form 160, 1 Aug 83, which may n[illegible]e used.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Decarburization. The cross section of the sample short and long loops shall be polished, etched with 3 to 5% Nital solution and examined under a microscope of not less than 500 power. Observation shall be made for compliance with the applicable drawing requirement.

4.5.2 Salt spray. Salt spray testing shall be conducted in accordance with ASTM Method B117. Observation shall be made for compliance with the applicable requirement in DOD-P-16232.

4.5.3 Hardness. Hardness testing shall be conducted in accordance with ASTM Method E18. Observation shall be made for compliance with the applicable drawing requirement.

4.5.4 Detent retention test. The sample links shall be tested using the test fixture specified herein. Links shall be assembled as specified on Drawing 9381628 and inserted in the test fixture. The load shall be applied to the mandrel as specified on Drawing 9381628. In performing this test, links shall be assembled so that each loop of each sample link shall be subjected to test. Observation shall be made for compliance with the applicable drawing requirement.

4.5.5 Tensile load test. The sample links shall be tested using the test fixture specified herein. The links shall be assembled into belts of five links each using steel mandrels or M51 dummy cartridges. Each belt shall be inserted in the test fixture and the tensile load shall be applied as specified on Drawing 9381628. Observation shall be made for compliance with the applicable drawing requirements. Upon completion of the test the sample links shall be scrapped.

4.5.6 Cartridge stripping test. The links shall be assembled to the test fixture specified herein and the applicable load shall be gradually applied at a uniform rate to that specified on Drawing 9381628. The retention force shall be recorded. Partial extraction shall be classified as a defect. Observation shall be made for compliance with the applicable drawing requirement.

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4.5.7 Free hinging test. The sample links shall be tested by assembling the links with standard 20mm dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid horizontally to its full length, with the open side of the links up, and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position. Observation shall be made for compliance with the applicable specification requirement.

4.5.8 Belt length test. The sample links shall be tested by assembling the links into belts of 25 links each using Government standard 20mm dummy cartridges or steel mandrels. One end of each belt shall be attached to a suitable hook on a horizontal table. To remove the slack from the belt the free end of the belt shall have a load of 115 pounds applied and then fully released. A load of 10 pounds shall then be applied and the center distance between the end links shall be measured.

4.5.9 Function test. A sample of 600 links taken from the first article shall be tested for functioning requirement (see 3.8) by assembling the links into belts using Government standard 20mm cartridges and firing through a 20mm, ASG-21 (B-52 Tail Gun) Fire Control System or equivalent in 3 bursts of 200 rounds each.

5. PACKAGING

5.1 Preservation and packaging. The links shall be preserved and packaged for shipment in accordance with Packaging Data Sheet 8448078 with levels as specified in the contract (see 6.2).

5.2 Packing. The links shall be packed in accordance with Packaging Data Sheet 8448078 with levels as specified in the contract (see 6.2).

5.3 Marking. Marking shall be in accordance with Packaging Data Sheet 8448078.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The M17A2 link is intended for use in the tail gun system of the B52 bomber.

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6.2. Ordering data. See MIL-A-48078.

6.3. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, Dover, NJ, 07801-5001, ATTN: AMSMC-QAF-I(D).

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.5 Definition of terms

6.5.1 Deformed link. A link whose shape or dimension or both have been mutilated or altered by processing to the extent that it is unsuitable for its intended use.

6.5.2 Cracks - tears. See Figures 1, 2 and 3.

6.5.2.1 Cracks - tears in principal areas. Any visible fracture of the link extending through the stock thickness in depth, or a distance in length equal to or greater than the stock thickness as depicted in Figure 1.

6.5.2.2 Cracks - tears in nonprincipal areas. Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length as depicted in Figure 2 and 3.

6.5.3 Burrs.

6.5.3.1 Burrs in principal areas. Any rough edge on any part of the link which, in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case or which would injure personnel in handling.

6.5.3.2 Burrs in nonprincipal areas. Any rough edge on any part of the link that does not fall within the description of 6.5.3.1.

6.5.4 Principal areas. Those inside and outside areas included by the bridge of the hinge and bridge area between nesting section and front gripping bands of the long loop; and those of the hinge brackets and guide lugs of the short loop (see Figure 4).

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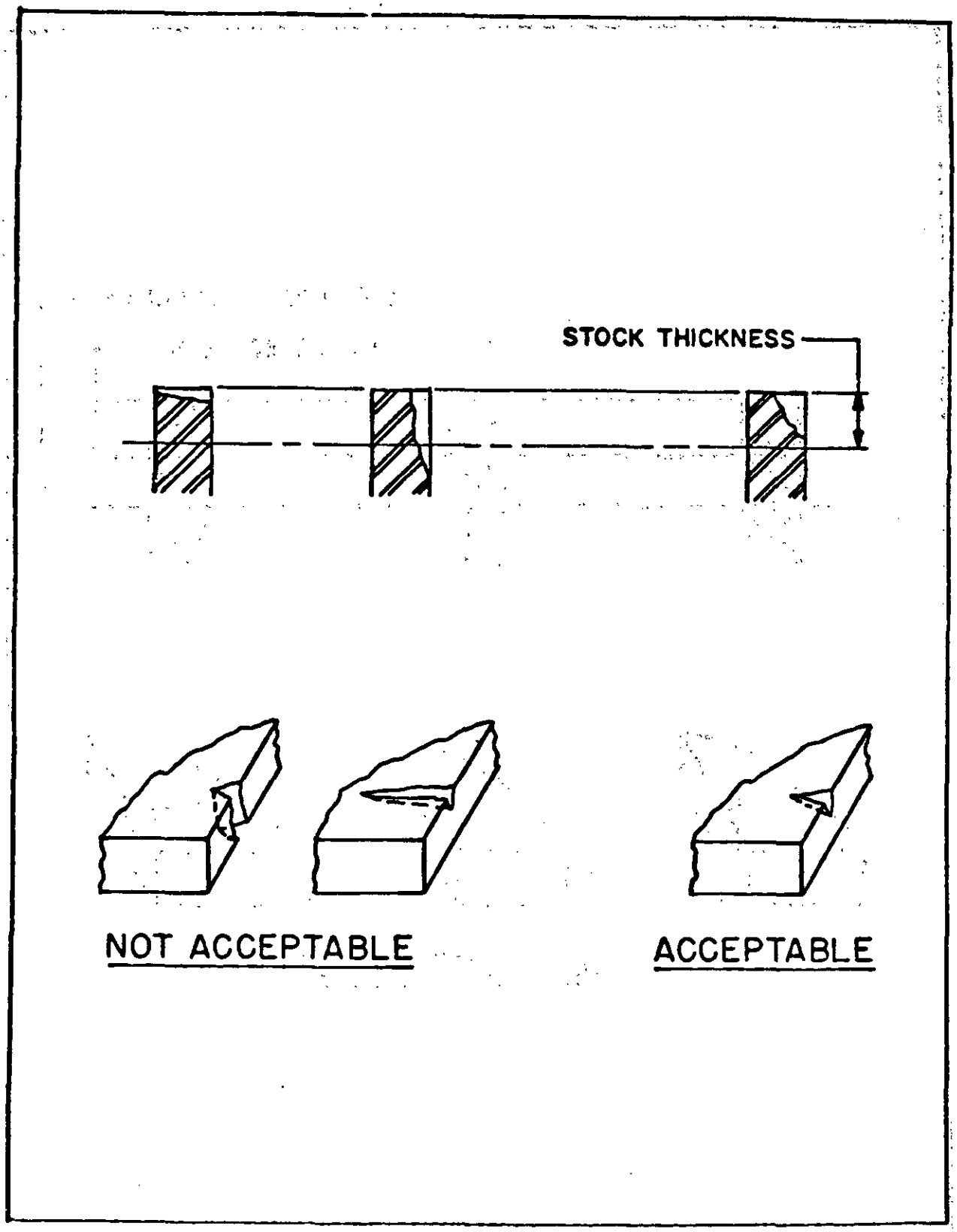


FIGURE 1

Cracks-tears in principal areas

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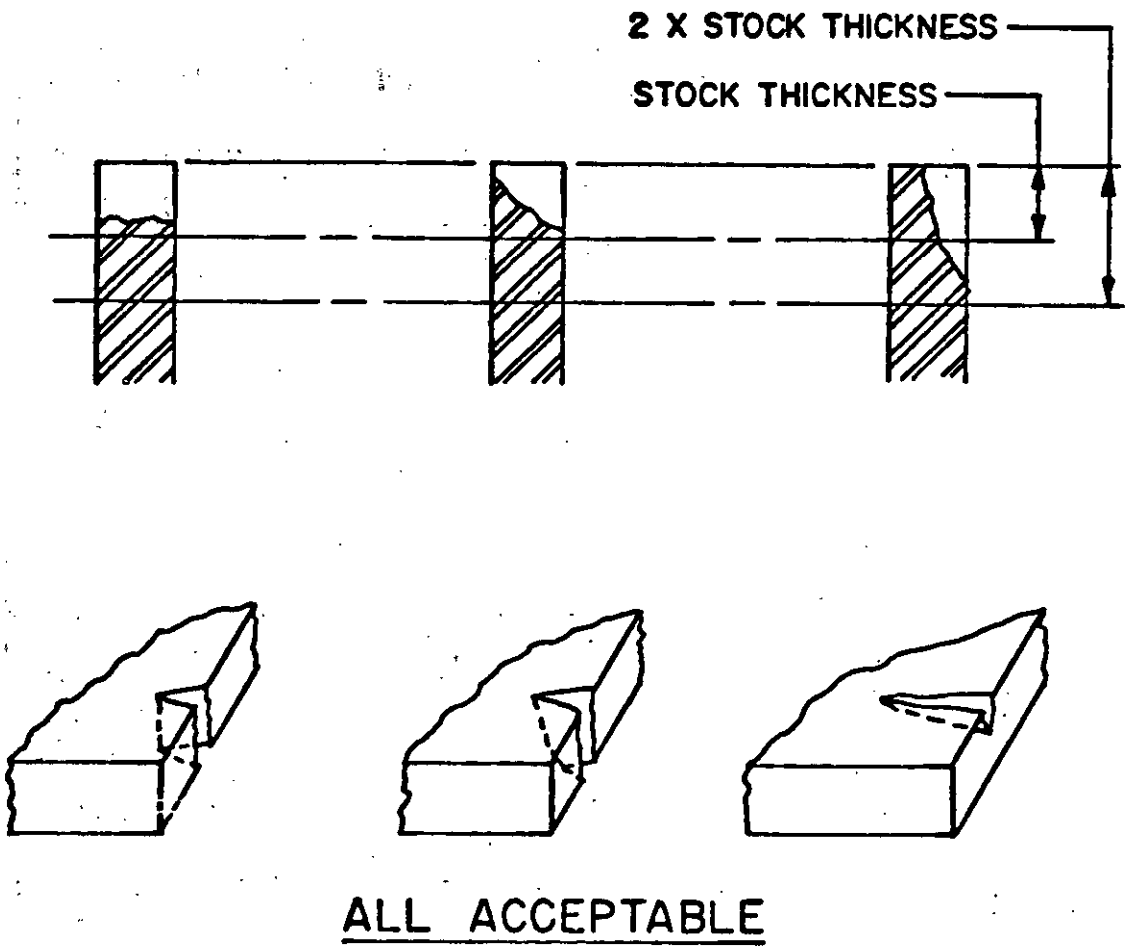


FIGURE 2

Cracks-tears in nonprincipal areas

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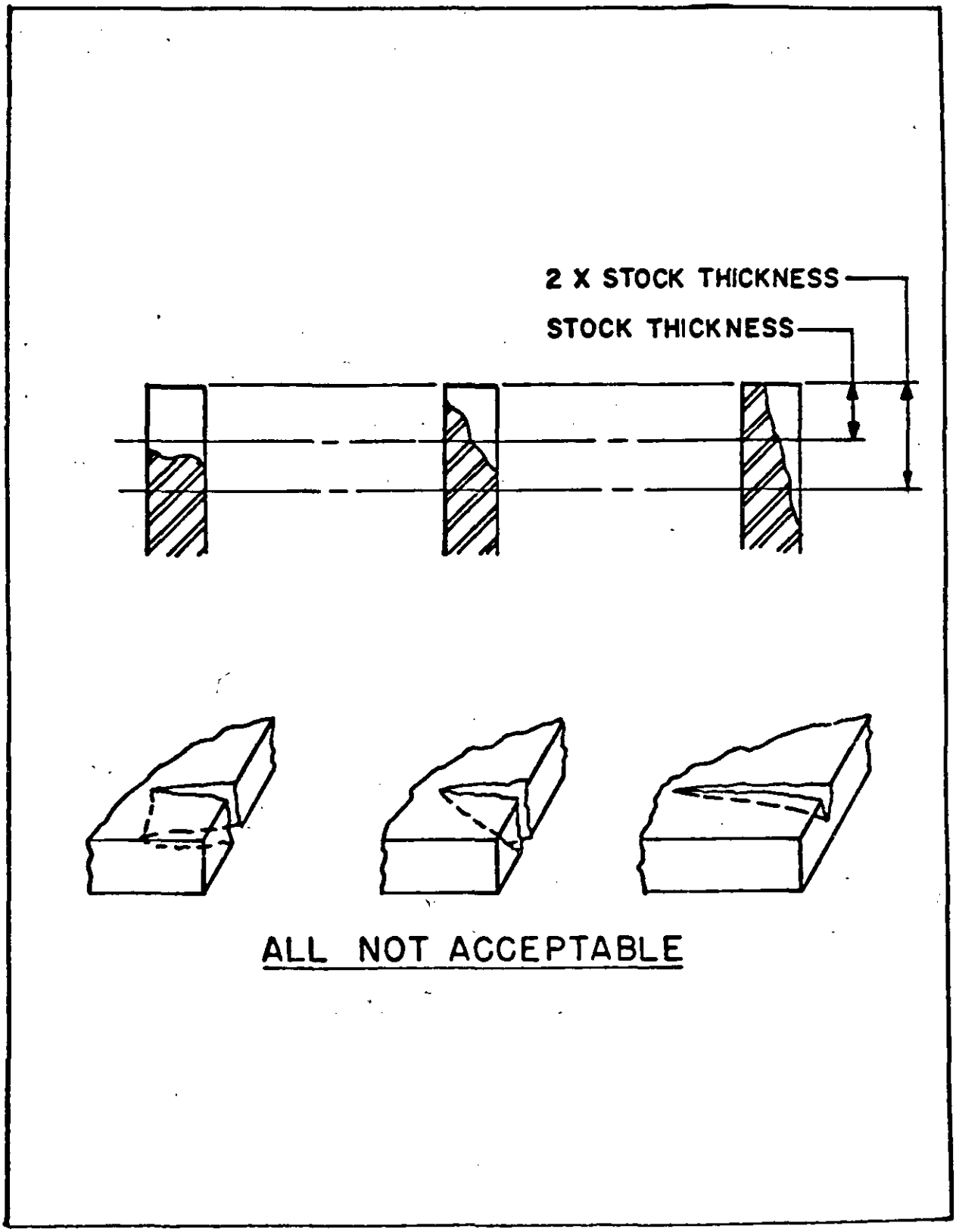
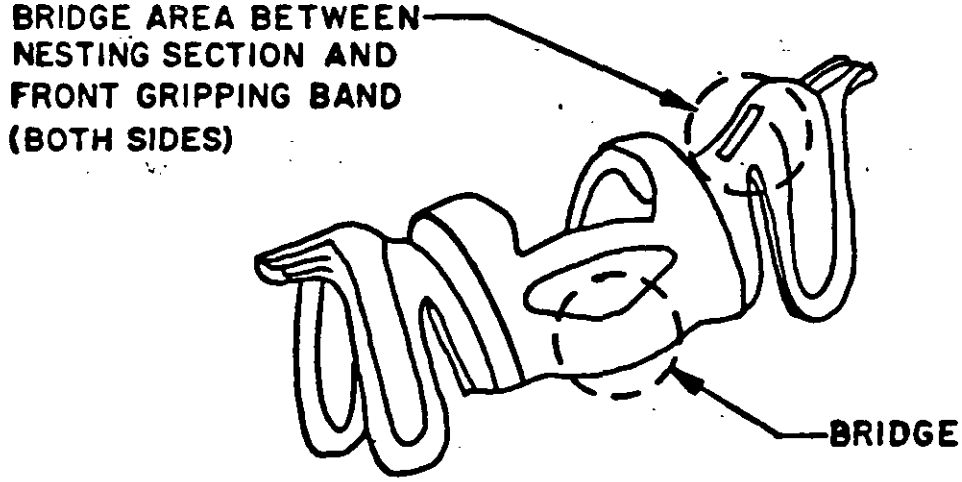


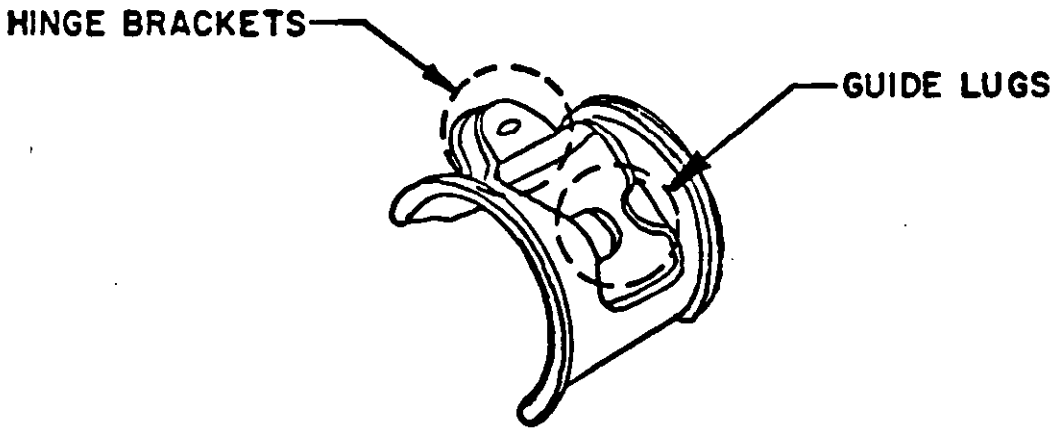
FIGURE 3

Cracks-tears in nonprincipal areas

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LONG LOOP



SHORT LOOP

FIGURE 4

Principal areas

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6.5.5 Partial extraction. When only one end of the mandrel or round becomes free.

6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 99

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Army - AR
Air Force

Preparing activity:
Army - AR

(Project 1305-0008)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L 45456E	LINK, CARTRIDGE, METALLIC BELT, 20MM: M17A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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